

Yaseen Mohamed

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Education

University of Waterloo, BAsc. Mechatronics Engineering Avg: **88.6%** | Excellent Standing Grad 2028

- **Relevant Coursework:** Embedded Systems/Microprocessors (C), OS & RTOS, Circuits, Actuators & Power Electronics, Sensors & Instrumentation, Signals & Systems, Data Structures & Algorithms (C++/C), MATLAB.

Skills

- **Embedded/Firmware:** C/C++, STM32 (HAL/LL), ARM Cortex-M, RTOS, DMA, timers/interrupts, PWM, Linux.
- **Interfaces/Test:** I2C, SPI, UART, JTAG/SWD, OpenOCD, CAN/CAN FD, Ethernet, Robot Framework.
- **Software/Data:** Python, Pandas, NumPy, Tkinter, Plotly, Pytest, SQL, Google BigQuery, Looker Studio, MATLAB.
- **Tools/Hardware:** Git, GitHub, GitLab, Azure DevOps, Jira, SSH, Altium, SolidWorks, 3D printing, oscilloscope, DMM, soldering.

Experience

System Integration Engineer

May - Aug 2026

Ford Motor Company

- Investigated large-scale vehicle software update issues using vehicle logs, **Google BigQuery**, **SQL**, and **Python** across 1M–80M-row datasets, validating edge cases and contributing evidence to JIRA tickets.
- Built 4 **Python** investigation tools using **Tkinter**, **Pandas**, and **Plotly**, plus various automation scripts, processing datasets containing 7M+ rows and reducing hands-on investigation time by **up to 90%**.
- Developed and optimized **BigQuery** queries using CTEs, window functions, partition filters, JSON parsing, and reusable tables, reducing runtime by **93%** and repeated retrieval time from **2 hours to 10–15 minutes**.
- Tested software-update failure scenarios on a cloud-connected vehicle integration bench, configuring ECU DIDs and cloud connectivity provisioning, flashing multiple software versions, testing OTA updates, and analyzing **CAN and Ethernet logs**.
- Traced **gateway ECU C++ code** to understand potential failure paths and created Markdown guides for AI-assisted code navigation, reducing investigations from **1–2 hours to 3–15 minutes**.

Embedded Software Engineering Intern

Sept - Dec 2025

Aversan Inc.

- Debugged and extended **STM32H7 embedded C firmware** for a safety-critical aerospace device, following **DO-178** processes, supporting bench debug and committing verified changes through the team's version control workflow.
- Built **5 Python tools** (PEP 8) over **UART** to parse and log test data, and automate firmware debugging/verification, uncovering tens of bugs and cutting test time by **70%**.
- Built a Linux Python GUI for **JTAG boundary-scan (OpenOCD)** to detect soldering defects, short circuits, and connectivity issues, and run **bit-banged I2C** tests without MCU firmware, catching manufacturing faults early and reducing rework.
- Implemented sensor/peripheral interfacing via GPIO bit-banged I2C on an STM32H7 using **DMA** and **HAL + custom timers**, plus **interrupt-driven UART RX** for status reporting, improving diagnostics and reducing component test time by **90%**.
- Created automated I/O tests in **Robot Framework** (via RIDE) on Linux-based **Automated Test Equipment (ATE)**, to verify set/read behavior and generate reports to be used for debugging, FAA certification, production, and maintenance.

Engineering Research Assistant

Jan - Apr 2025

University of Waterloo School of Optometry and Vision Science

- Built a pressure-sensing footstep detection system using an **Arduino**, **48-channel MUX**, **custom PCB**, and **Raspberry Pi**, and developed a decay-compensated **C++/Python** acquisition pipeline that increased sampling frequency by **99%**.
- Built a **Python GUI** with a **grouping algorithm** to preprocess and visualize foot-pressure data, and designed **SolidWorks/FDM** equipment upgrades that reduced setup time by **80%**.

Projects

Quadcopter Prototype | [GitHub](#) · [Demo](#) | C++, Arduino, IMU, timers, Control (PID)

June - Aug 2020

- Built an Arduino quadcopter prototype with **PID stabilization** in **C++**, using **IMU (gyro) feedback** and custom **PCB/wiring** for real-time attitude control and tuning.

Obstacle-Aware Arduino Robot | [GitHub](#) · [Demo](#) | C++, Arduino, Real-time control

Oct - Dec 2023

- Built an obstacle-aware robot using **Arduino C++**, with an **LCD UI**, **ultrasonic ranging**, and button-controlled 2-state **finite-state machine** (idle/run) with debounce, in a simple real-time control loop for obstacle avoidance.